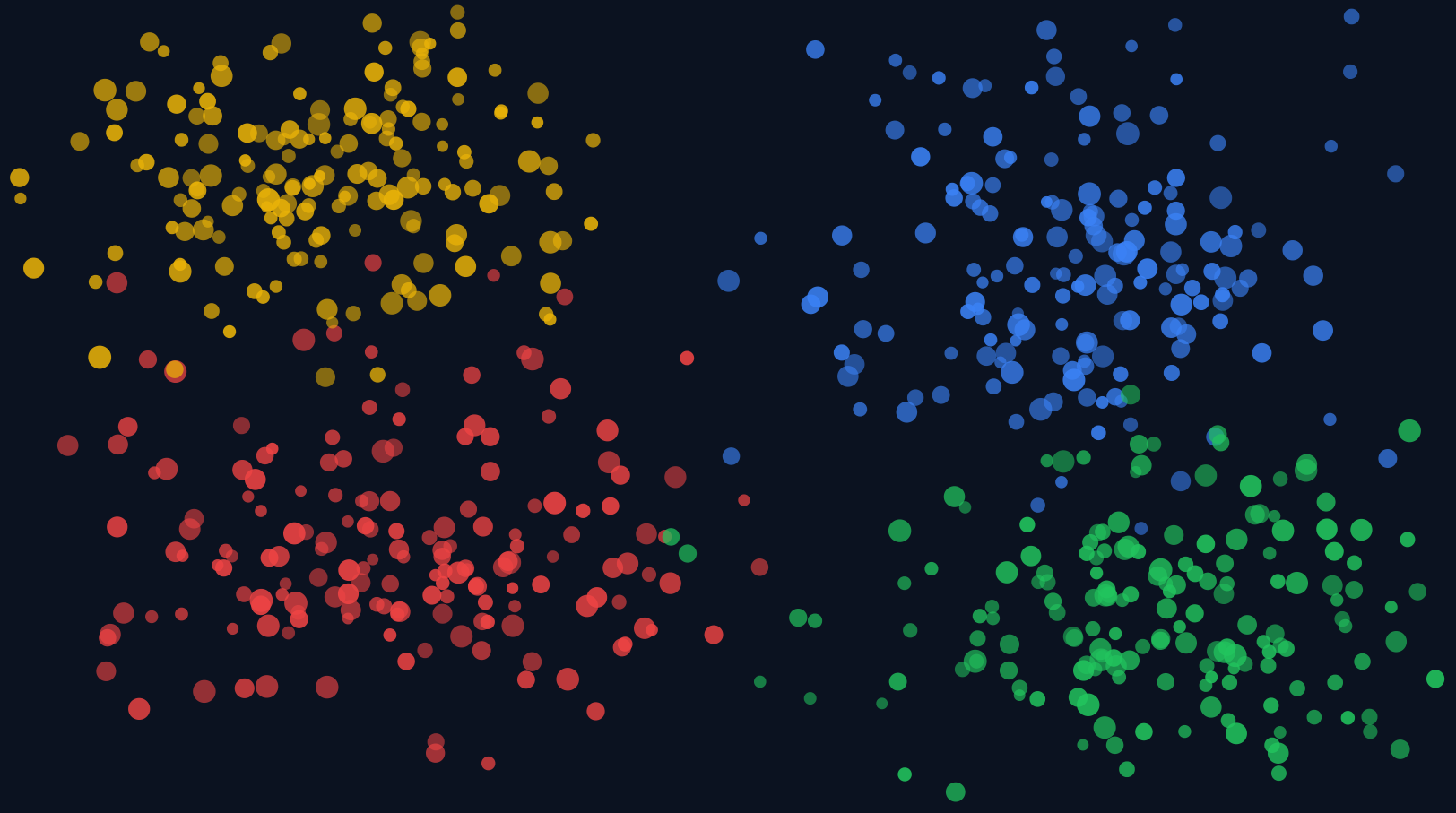
UNSUPERVISED MACHINE LEARNING · PROJECT

I grouped 2,237 customers into 4 segments, with no labels.

SmartCart customer segmentation using PCA, K-Means and Hierarchical clustering in Python.



Rohan Rathod

Swipe to see every step →

THE PROBLEM

One campaign for everyone wastes money.

SmartCart has customer data but no ready-made groups. Can the data reveal its own natural segments?

2,240	22	0
customers	raw features	labels

Clean → Engineer → Scale → PCA → Cluster → Profile

STEP 1 · CLEAN + ENGINEER

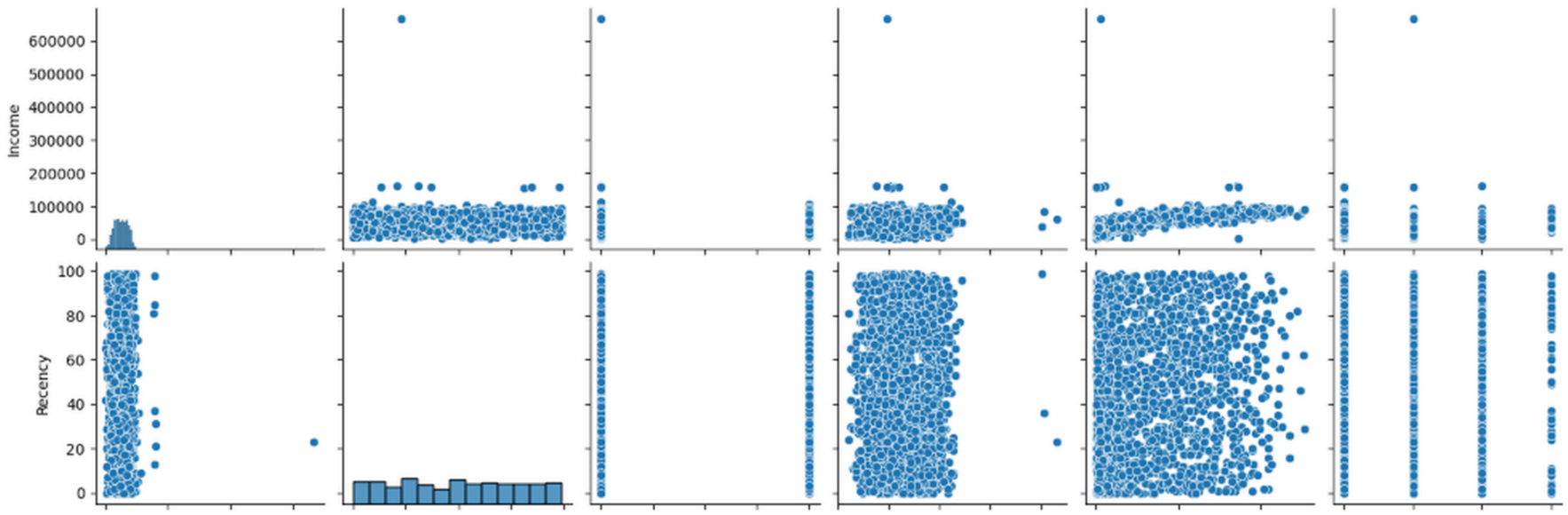
Turn raw columns into signals.

- **Missing Income** (24 rows) filled with the median
- **Age** from birth year
- **Customer_Tenure_Days** from sign-up date
- **Total_Spending** = wine + fruit + meat + fish + sweets + gold
- **Total_Children** = kids + teens at home
- **Education** 5 levels → 3 (Under / Grad / Post-grad)
- **Marital status** → Living with Partner or Alone

22 columns in → **15** columns after dropping redundant ones

STEP 2 · OUTLIERS

Pair-plots expose the weird points.

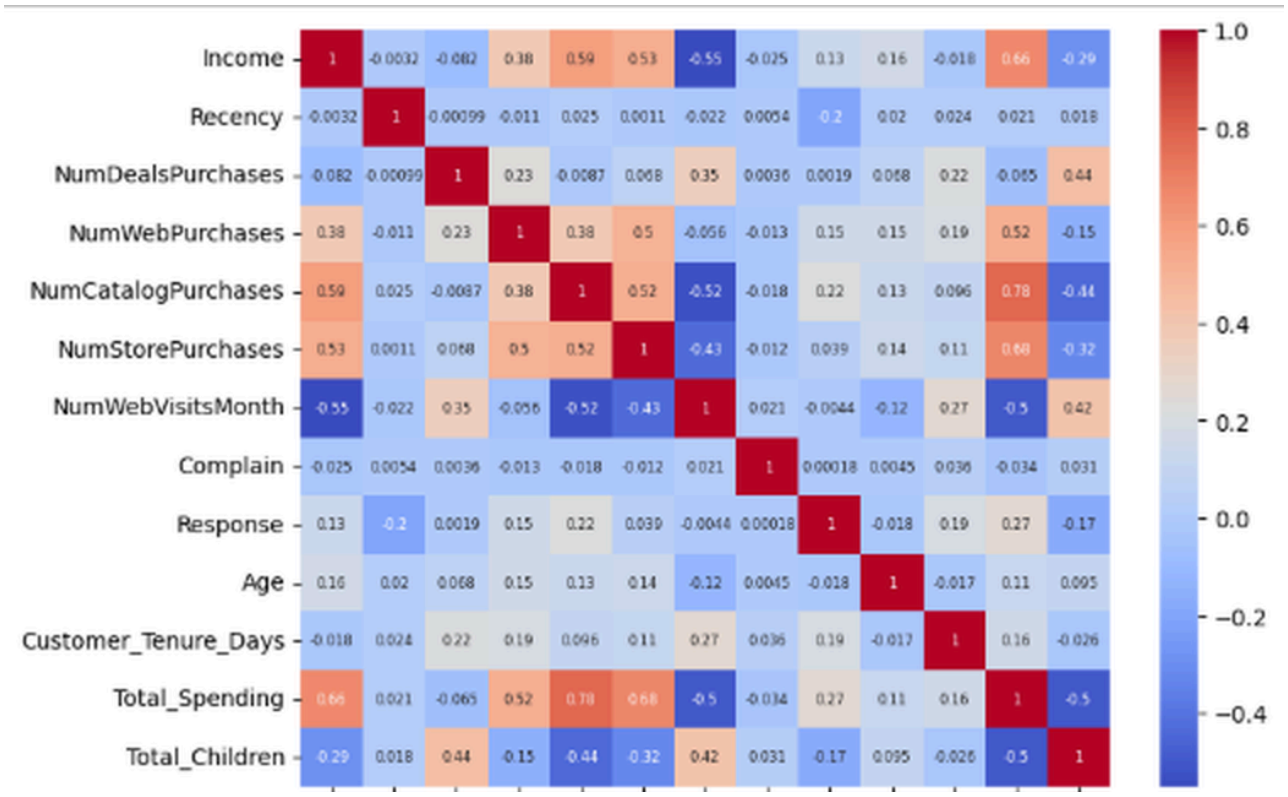


Relative plots of Income, Recency, Response, Age, Spending, Children

Extreme values stand out immediately. Unrealistic ages (90 and above) were filtered: **2,240 → 2,237 customers**.

STEP 3 · CORRELATIONS

What moves together?



Correlation heatmap of numeric features

0.78
Catalog purchases ↔ Total spending

0.66
Income ↔ Total spending

−0.55
Income ↔ Web visits / month

−0.50
Children ↔ Total spending

Income and Total_Spending became my lens for reading the clusters later.

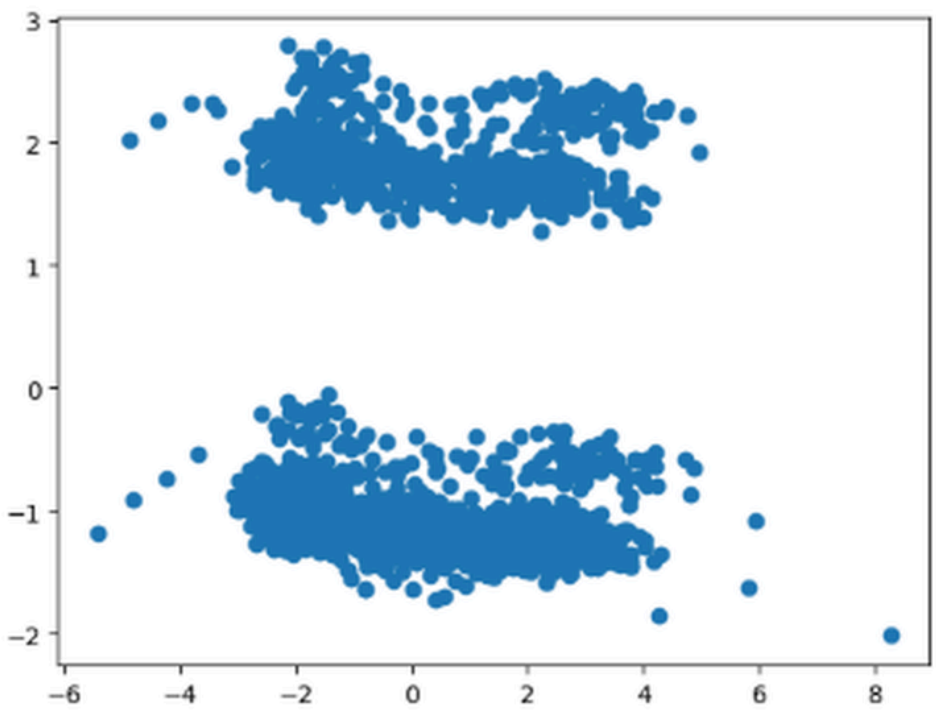
STEP 4 · ENCODE → SCALE → PCA

Prepare, then shrink.

One-Hot Encoding
Education + Living_With → 18 numeric features

StandardScaler
So Income in the tens of thousands doesn't drown Recency

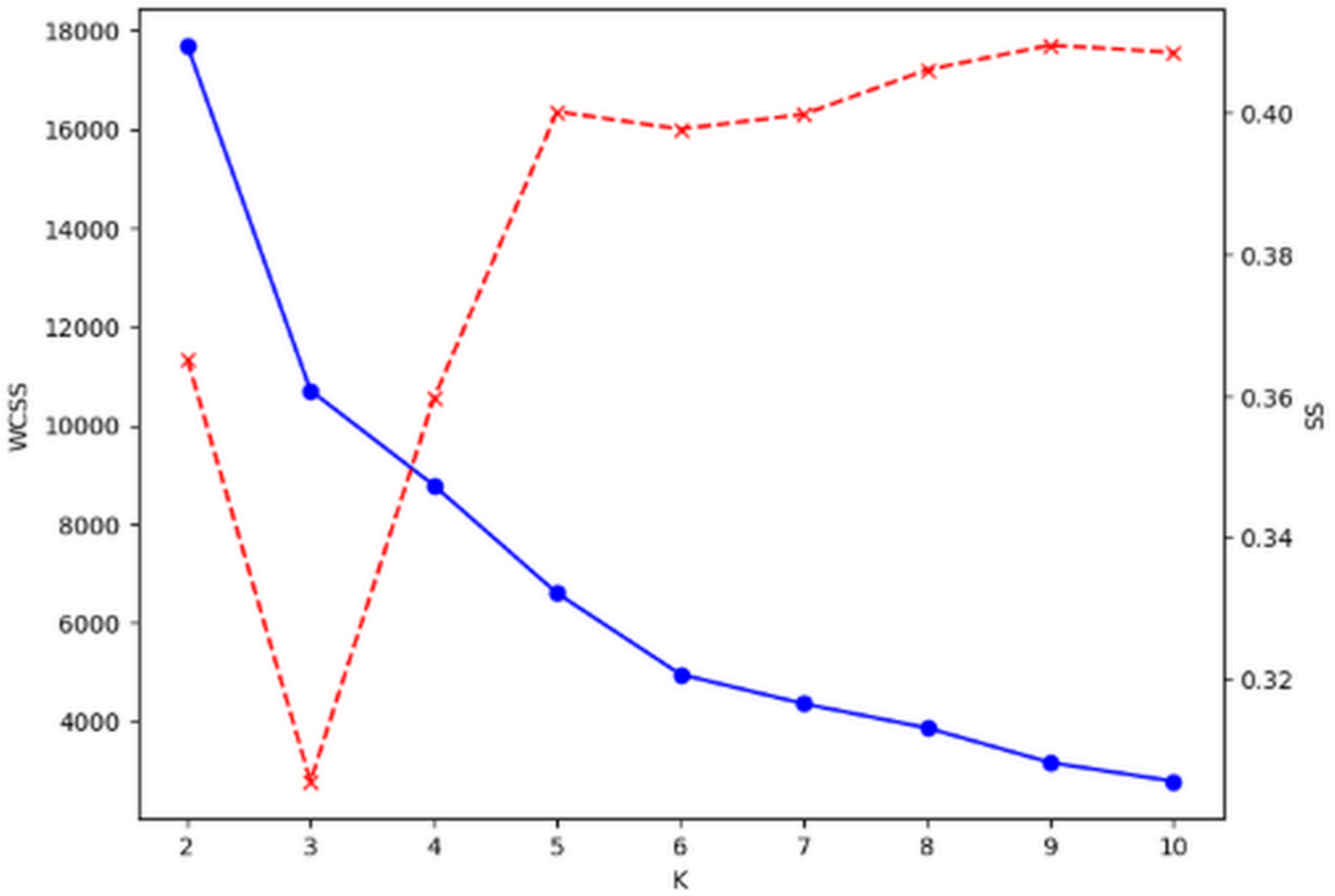
PCA → 3 components
22% + 11% + 10% = ~44% of variance, enough to visualize



Data projected onto the first two principal components

STEP 5 · CHOOSING K

How many clusters?

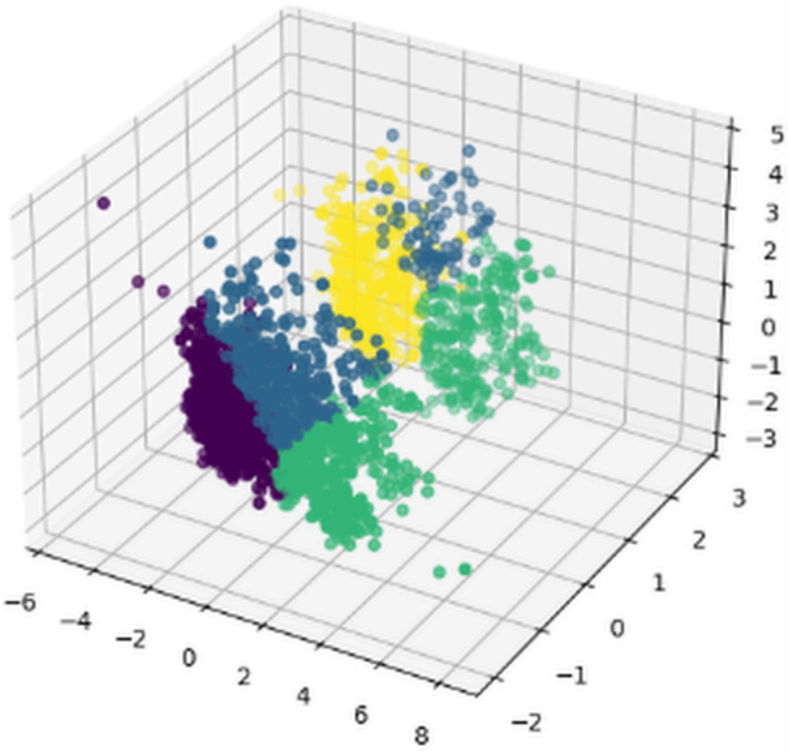


WCSS (blue) and silhouette score (red dashed) for K = 2 to 10

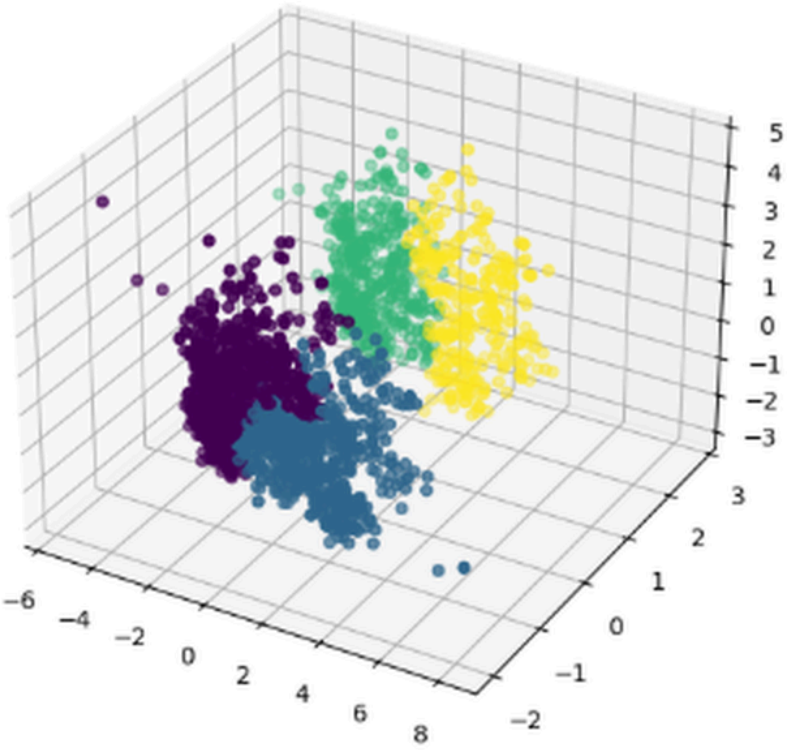
- **Elbow method** (auto-detected with KneeLocator): **k = 4**
- **Silhouette** ≈ 0.36 at k = 4, rising to ≈ 0.40 beyond k = 5. A trade-off between score and segments you can act on

STEP 6 · CLUSTERING

Two algorithms, same k = 4.



K-Means

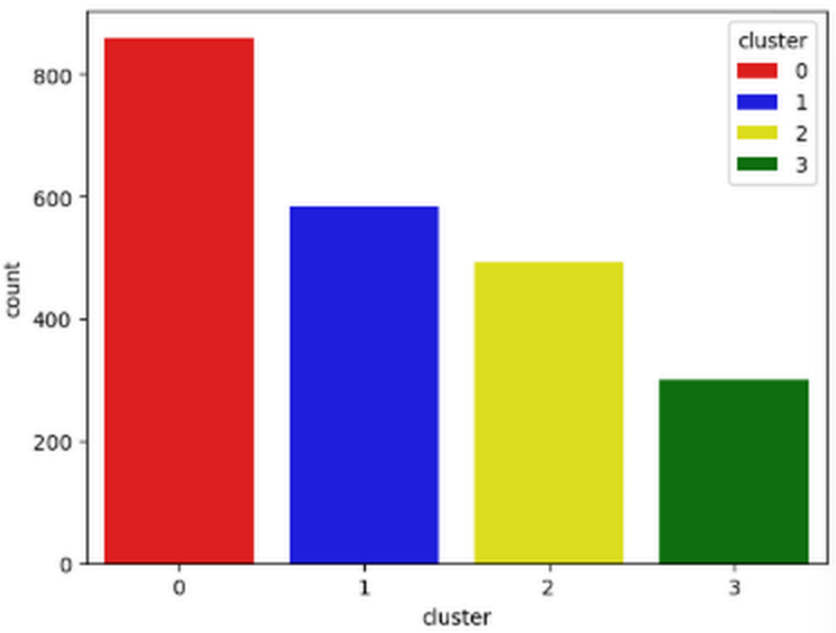


Agglomerative (Ward linkage)

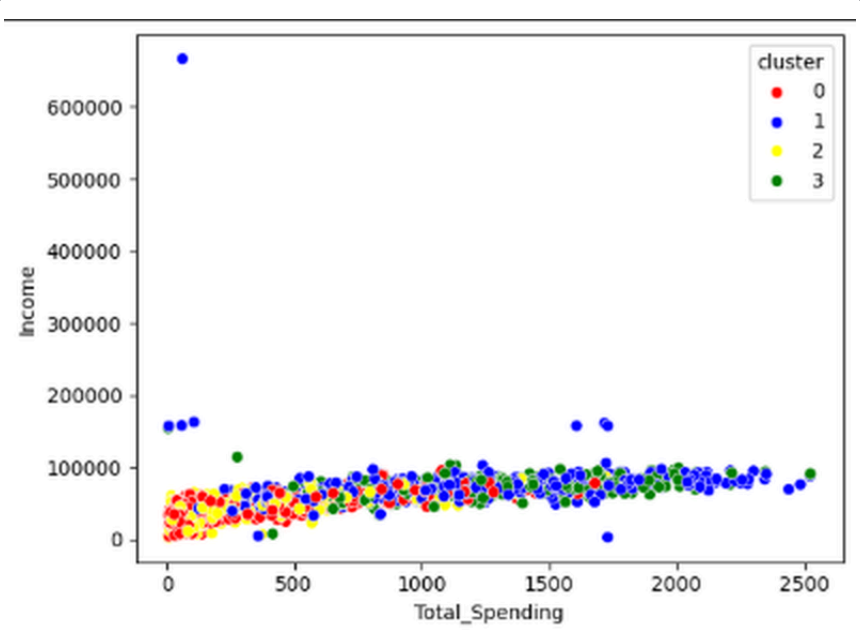
Both plotted in 3D PCA space. I used the **hierarchical labels** to profile each segment.

STEP 7 · LOOKING AT THE CLUSTERS

Sizes, and the income vs spend pattern.



Customers per cluster: 860 · 583 · 494 · 300



Income vs Total_Spending by cluster

Spending rises with income, but the clusters split **low spenders** from **high spenders** at similar incomes.

THE RESULT

Meet the 4 segments.

Affluent solos

Green · lives alone · 300 customers

72.1K	1,258	0.4
avg income	avg spend	children
1.7	3.4	37.7%
deal buys	web visits/mo	responded

Affluent couples

Blue · with partner · 583 customers

70.4K	1,098	0.5
avg income	avg spend	children
1.9	3.7	13.4%
deal buys	web visits/mo	responded

Budget families

Red · with partner · 860 customers

40.1K	258	1.3
avg income	avg spend	children
2.7	6.5	9.3%
deal buys	web visits/mo	responded

Budget solos

Yellow · lives alone · 494 customers

39.9K	235	1.2
avg income	avg spend	children
2.6	6.4	12.8%
deal buys	web visits/mo	responded

Segment names are my own interpretation of each cluster's averages.

SO WHAT?

What the business can do.

Affluent solos

Campaign response of **37.7%**, roughly 3–4× the other groups. Prioritize them for targeted offers.

Affluent couples

High spend (~1,100) and heavy catalog and store buyers. Good fit for loyalty perks and bundles.

Budget families + solos

Spend 4–5× less, buy more on deals and visit the website ~6.4× a month. Try coupons and web promos.

These are patterns in the data, not proof of cause. An A/B test would validate them.

WHAT I TOOK AWAY

Learning in public.

Learned

- Feature engineering mattered more than raw columns
- 3 PCA components keep only ~44% of variance
- Elbow and silhouette don't always agree

Next

- Tighter outlier handling (a few high incomes remain)
- More PCA components, plus GMM / DBSCAN
- Validate segments with an A/B test

Code and notebook on GitHub

github.com/RohanRathod81

Open to Data Science opportunities. Which clustering algorithm do you reach for first? Tell me below 👇